



■ By M.G.K. Menon

Since Independence, India has built many institutions that it can be truly proud of. These institutions essentially represent the implementation of a vision, often of inspiring individuals like Homi Bhabha, Vikram Sarabhai, Jawaharlal Nehru and J.R.D. Tata. All of them were deeply conscious that the needs of engineering and scientific research demanded completely different organisational and administrative patterns compared to what was considered normal in a static society, particularly under colonial rule. Bhabha basically followed three principles: the first was, "official support from government need not entail government control". The second was that in all the organisations he built up, whenever he found a person in whose abilities he had full confidence, he extended all the support he could. The third principle he adopted was: in a developing nation where skilled manpower is scarce, a project must be executed with the maximum number of skillful people (so as to provide training opportunities and not the minimum number which financial authorities usually do). For him as well as for Sarabhai, administration was supposed to play the role of a supporting system and not of an overpowering controller.

Even before Independence, there were several institution builders, whose dreams crossed borders. One of them was Jamsetji Tata. He created one of the country's finest educational post-graduate institutions—the Indian Institute of Science in Bangalore.

Against all odds, he was also able to set up the largest steel plant in the British Empire during the First World War. He was an extraordinary man who understood the significance of the industrial revolution in the West but also wanted a prosperous India. Like M. Visvesvaraya in the old Mysore state, he generated clean and environmentally sustainable hydro-electricity. The house of Tatas also gave us the Sir Dorabji Tata Trust, which was instrumental in setting up institutions like the Tata Institute of Fundamental Research (TIFR), Tata Institute of Social Sciences, Tata Memorial Hospital—one of the finest for treatment and research in cancer—and the National Centre for Performing Arts. Then there was Mahendra Lal Sarkar, who set up the Indian Association for Cultivation of Science (IACS) modelled on the Royal Institution in London. It was at IACS that Sir C.V. Raman

did his Nobel Prize-winning research work in physics.

So, how did these great men build the foundations on which the great edifice of modern India has been built? Each institution has a history, which takes it outside the brick and mortar of physical space. Take the example of how TIFR was set up. Bhabha had returned to India after many years of working in Cambridge University, which had given him an opportunity to associate with some of the greatest names in science. He had also acquired a deep understanding of nuclear physics, and of the possibilities inherent in it. Because of the Second World War, he could not go back to Europe and had to continue his research in Bangalore. Bhabha noticed a lull in the publication of papers in nuclear physics in the 1940s. At the same time, he was also able to identify with his country. He realised that for the country to develop, it had to give modern science its due and develop an economy based on modern technology. Bhabha was also of the view that energy was essential for development. All these thoughts came together in his vision.

In 1943, Bhabha wrote a letter to J.R.D. Tata proposing the setting up of a new institution, which later came to be called TIFR. Encouraged by a favourable response, he made a formal proposal to the Sir Dorabji Tata Trust. In this letter, he said: "Moreover, when nuclear energy has been successfully applied for power production, in say a couple of decades from now, India will not have to look abroad for its experts but will find them ready at hand."

This was Bhabha's vision of abundant economic nuclear power, which he felt was vital for the development of India. He wrote this in Bangalore in 1944, more than a year before Hiroshima. Work on the atom bomb was being carried out with the greatest secrecy in the West.

Sarabhai, like Bhabha, was a great institution builder. A scientist, he set up an institution, the Physical Research Laboratory, where he could devote himself to his basic love for fundamental research. Since his family was connected with the textile industry, he realised the importance of research conducted by industry and for industry. He established the Ahmedabad Textile Industrial Research Association and was its honorary director. In pursuit of his scientific interest, he used to travel abroad quite often particularly as a visiting professor at the

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